PRESENSEA: PRESSURE SENSOR BASED SEA MONITORING · MOBILITAS PLUSS MOB3JD1212

Listening to the Baltic Sea

Non-Invasive Vessel and Marine Acoustic Monitoring with pressure sensors, acoustics and DAS

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THE JOURNEY

2020 — First MSCA Application
Applied for MSCA Individual Fellowship

Dec 2023 — PreSenSea Funded
Mobilitas Pluss MOBJD

May 2025 — Project Starts
Researcher joins Centre for Biorobotics, TalTech

May 2026 — Year 1 Complete
Results presented at this poster

THE PROBLEM

The Baltic Sea is the acoustic equivalent of a **cocktail party gone wrong**: dense shipping traffic fills the low-frequency band with continuous noise — precisely where many marine species produce their vocalisations. Understanding the effects of anthropogenic noise on marine life is crucial for biodiversity preservation in the face of other pressures, such as climate change, pollution and habitat loss.

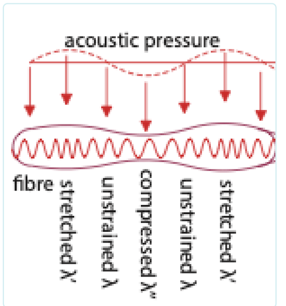
Current fish stock monitoring relies on **trawl surveys**: invasive, expensive, and unable to capture reproductive behaviour. A detection system based on fish vocalisations could provide marine conservationists and authorities with timely, non-invasive monitoring data.

PUBLICATION **TalTech Trialoog**

"Läänemeri ootab laineid lõövat seiret" — science communication article in TalTech's public magazine, presenting the project's mission to the general public and attracting new collaborators and students.

trialoog.taltech.ee

WHAT IS DAS?



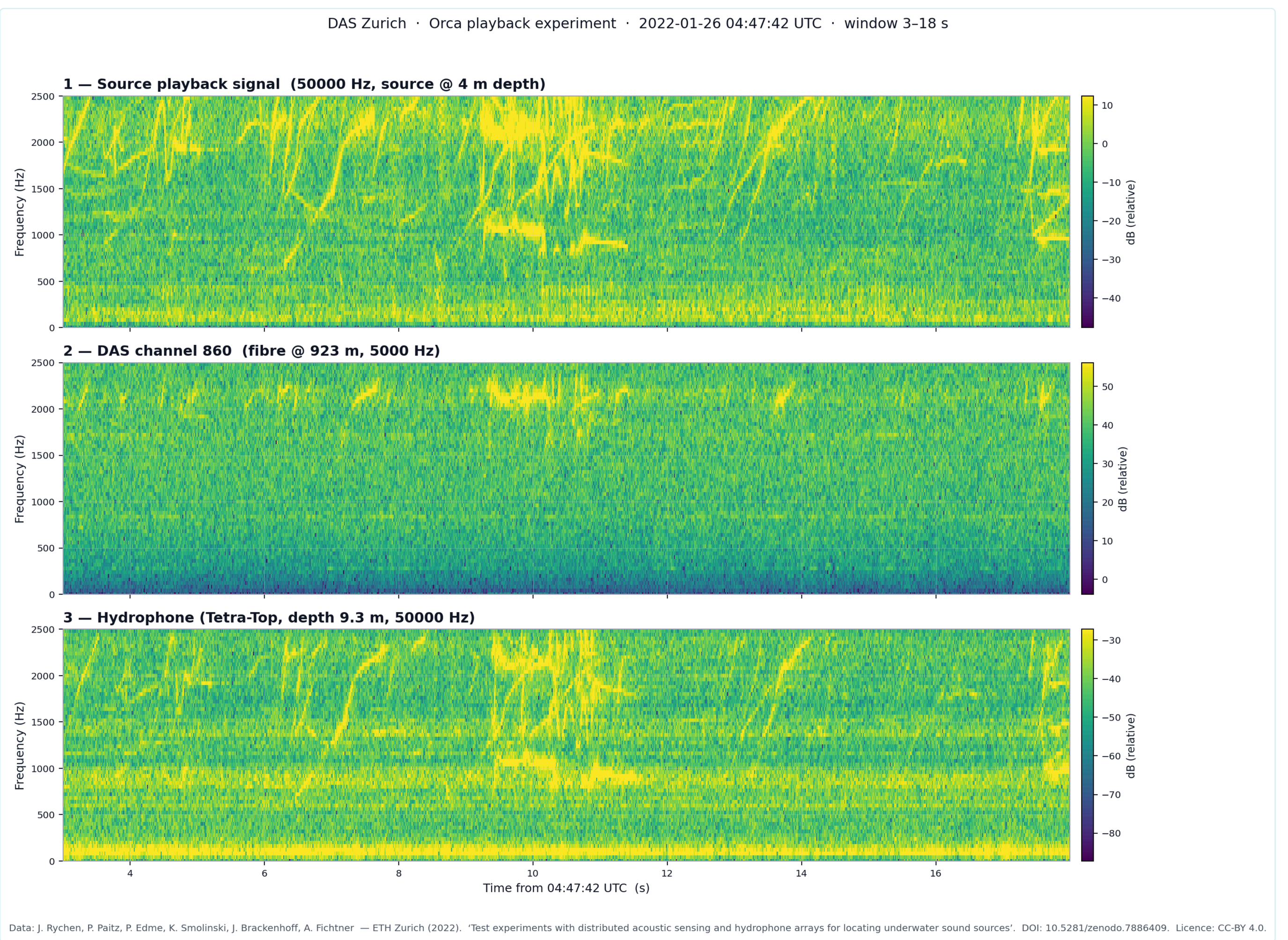
Distributed Acoustic Sensing uses Rayleigh backscattering along standard telecom fibre to create **thousands of phase-coherent virtual sensors** over 100+ km.

This transforms existing cable infrastructure into a **passive array** — no batteries, no retrieval, no maintenance.

Key challenges include enormous data volumes, telecom restrictions, limited bandwidth and coupling losses.

DAS for Fish Detection — Evidence

Missale, Ravasi et al. (2025) deployed DAS alongside a hydrophone on a Red Sea reef and detected **fish vocalisations** (100–1000 Hz) and snapping shrimp choruses with diel patterns matching biological rhythms.



Data: J. Rychen, P. Paitz, P. Edme, K. Smolinski, J. Brackenhoff, A. Fichtner — ETH Zurich (2022). 'Test experiments with distributed acoustic sensing and hydrophone arrays for locating underwater sound sources'. DOI: 10.5281/zenodo.7886409. Licence: CC-BY 4.0.

Spectrograms generated using BSc Thesis: Multiscale Underwater Acoustic Event Analysis

→ If transferred to Baltic cod spawning grounds, this changes everything.

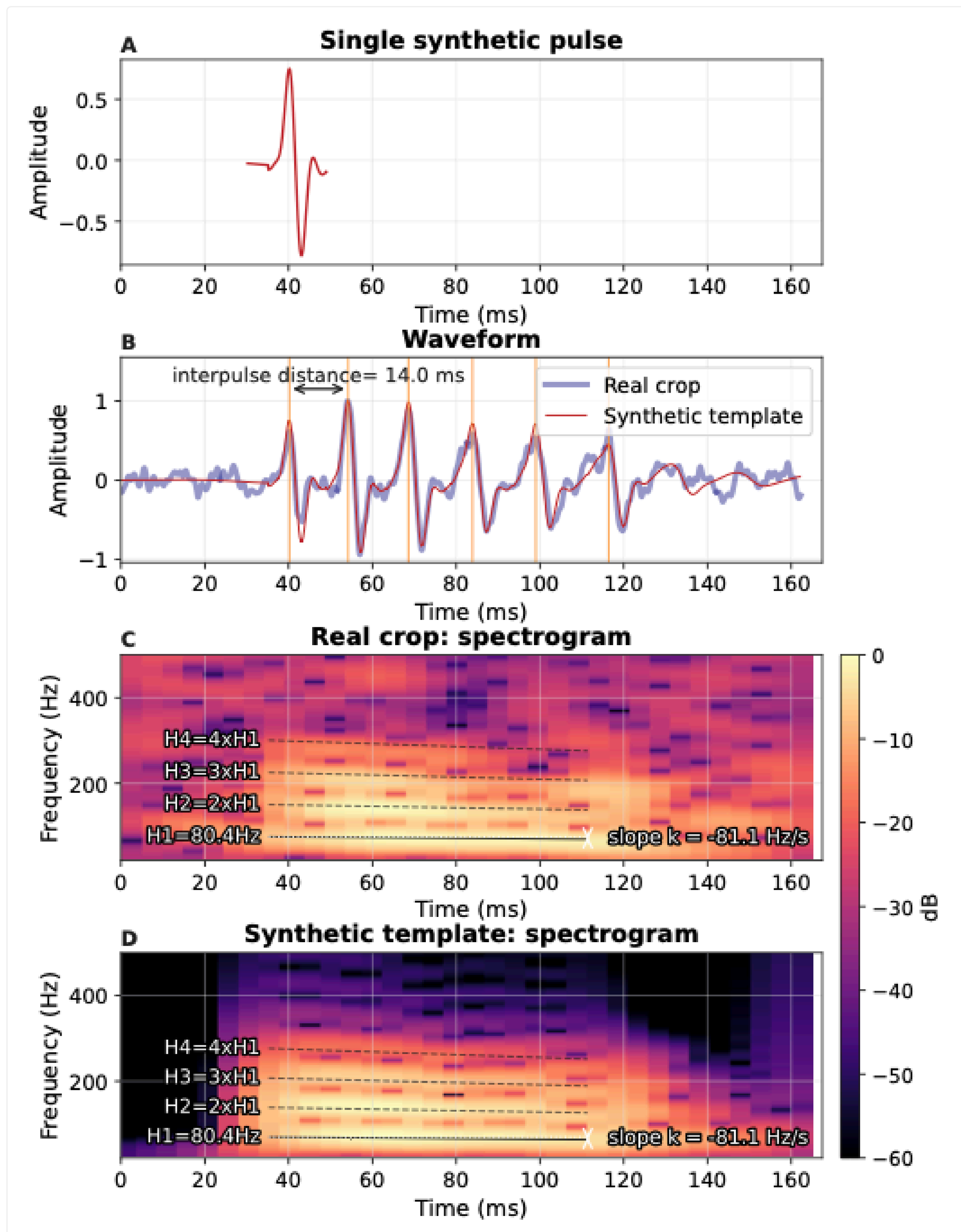
Existing submarine cables and sensors can become passive, continuous marine monitoring arrays — detecting vessels, sea state, and potentially marine life using pressure sensing and acoustics.

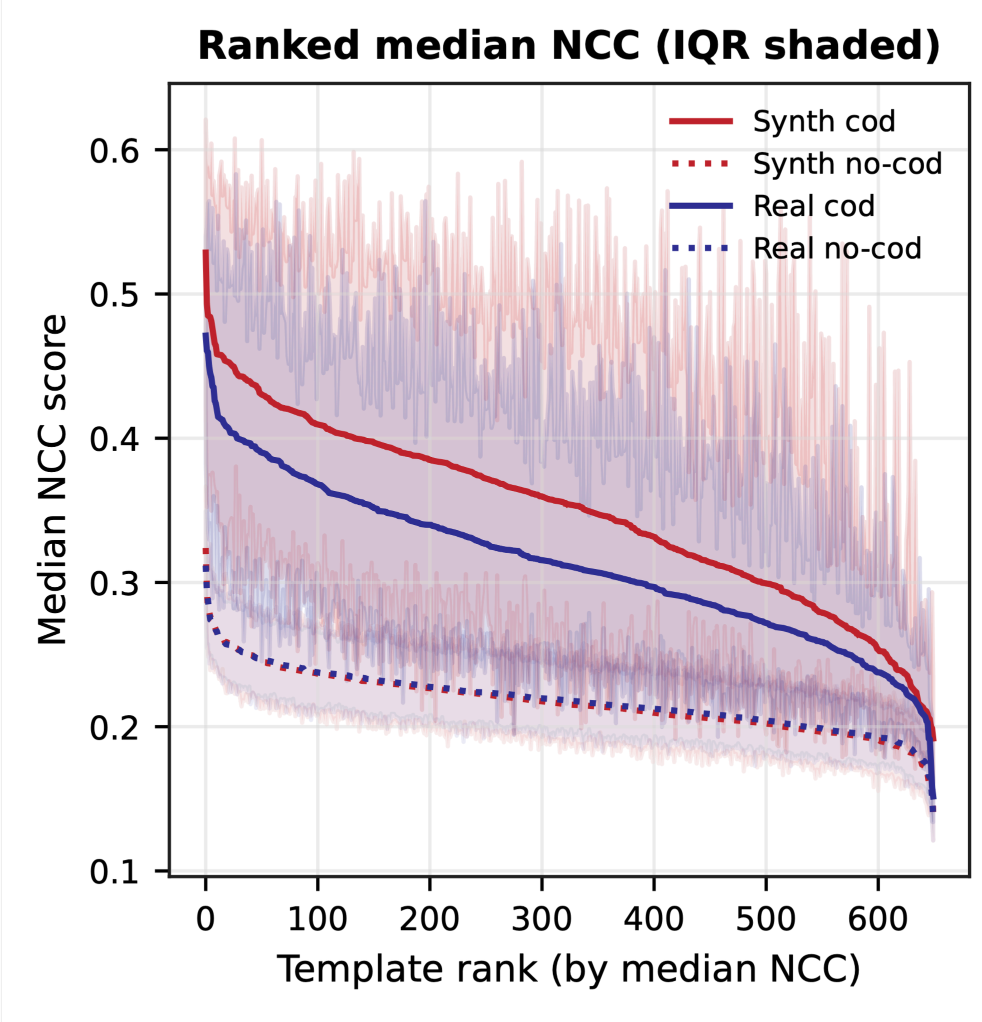
BNAM 2026 **Synthetic Cod Grunts**

Bio-inspired parametric grunt generator replaces station-specific templates. Modelled as pulse train of Gaussian-windowed swept harmonics, calibrated on 1,348 cod grunts from NOAA SanctSound. Alone without ML achieved F1 = 0.975 detection rate in presence of shipping noise and other biological sources. It can be further improved using other features, e.g. from foundational models.

F1 = 0.975

Up to 97.5% detection of cod vocalization
in presence of shipping noise and other biological sounds





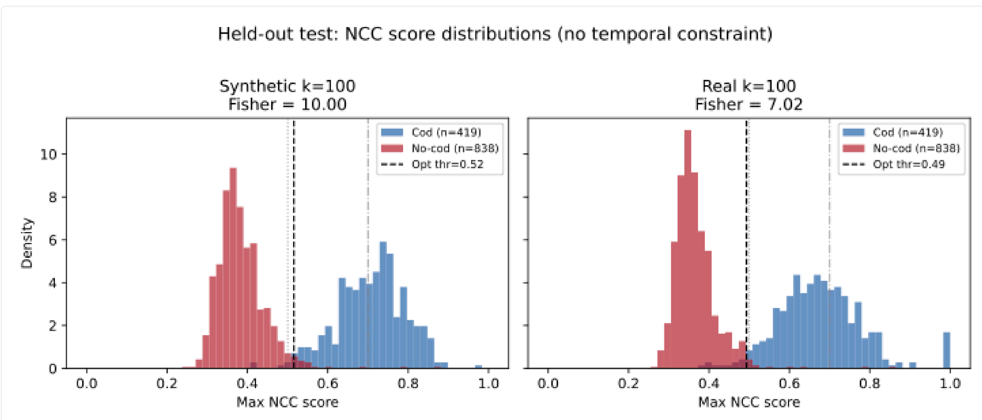


Fig 1. Single Template NCC, Detection Score Distributions, and Real vs Synthetic Template Pair

MSC THESIS **Anna Bass (2026)**

"Detection of Atlantic Cod Vocalizations in Underwater Passive Acoustic Monitoring Data under Label Scarcity"

Evaluated **SurfPerch** foundation model embeddings + active learning for cod grunt detection under label scarcity. Key finding: window-target alignment (1s vs 5s) critically affects embedding quality.

TOOL **DAS Audit**

Open-source script generating **interactive HTML reports** from sanity checks of large DAS datasets. Optimised for cloud / big data / low RAM. Works across different equipment and datasets.

github.com/presensea/DAS

Available upon request · TODO: implement more SOTA methods

THE VISION

Current Baltic cod monitoring is **invasive, expensive, and failing** to capture reproductive state. DAS could turn **thousands of km of existing submarine cables** into a continuous, passive observation network — replacing seasonal trawl surveys with year-round data.

FUTURE WORK

ReSound

Non-Invasive Metric for Cod Spawning Function. BiodivConnect Stage 2: DAS + PAM across 4 sites (Bornholm, Danish Straits, Mecklenburg, Svalbard). Multi-stock acoustic dialect characterisation.

Partners: NTNU · SDU · Thünen Institute

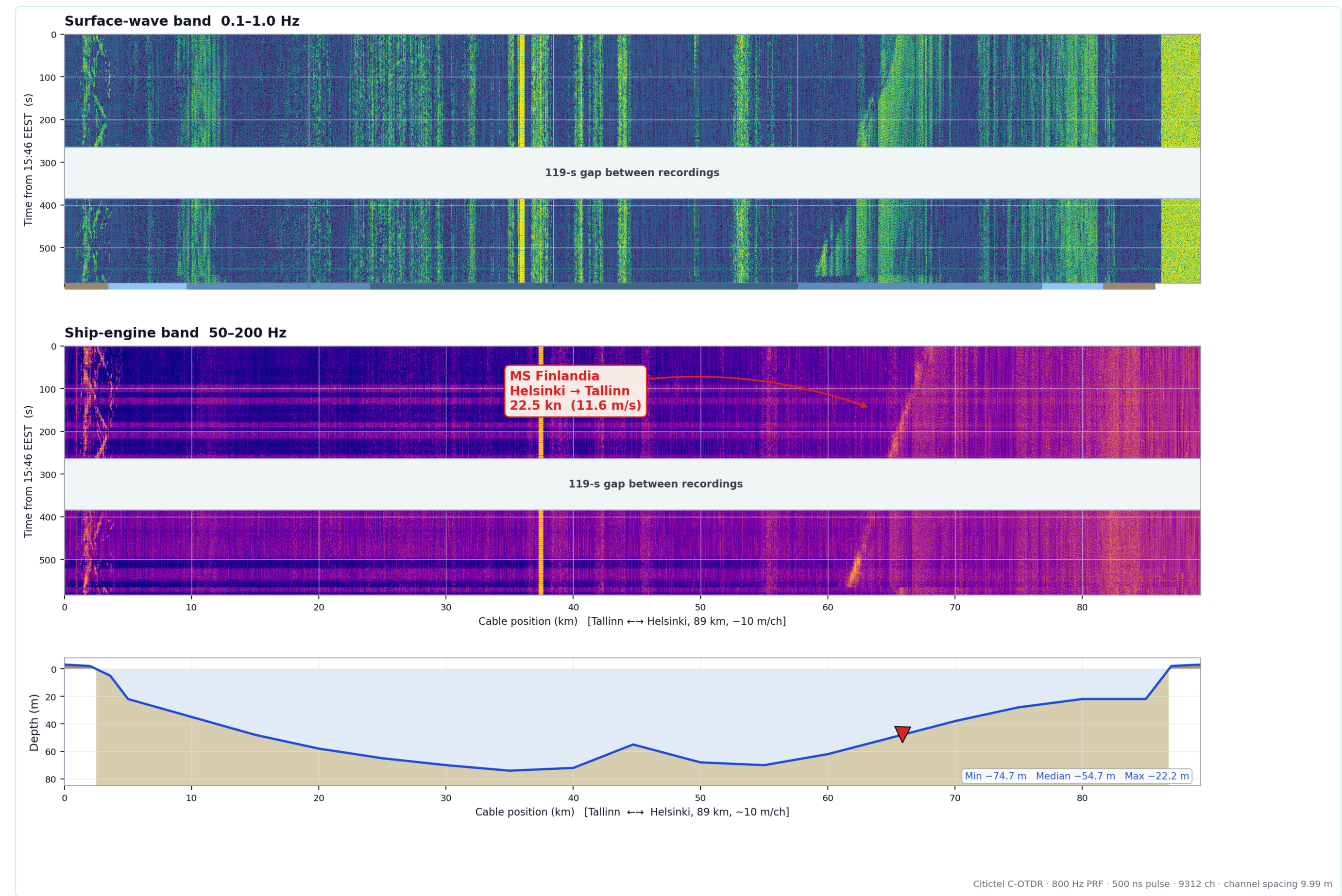
Wellamo / SEAGUARD

Fish-Sensing Hydrophone combining Hydromast flow sensors with hydrophones for noise-cancelled detection at Baltic Sea wrecks. Testing if metal artificial reefs serve as cod spawning grounds.

DAS Sea State & Ship Wakes

First submarine DAS dataset in Estonia. Detecting **ferry transits** and sea states along the 100km Tallinn-Helsinki cable.

Kaida Kaeval. The work has been funded by the European Commission, grant agreements 10113933 (ECO-eNET) and 101189703 (ICON).



In Progress & Future Proposals

- OSAW (PSG Application): Fusing vector sensor data with DAS for directional acoustic monitoring.
- NAUTILUS: Framework for rapidly deployable edge-computing aquatic sensor networks.
- Hydromasts Dataset: Processing and extracting continuous flow velocity parameters.
- DAS Consilium & Zurich Audit: Interactive dashboards and sanity tools for large datasets.
- Local Inference: Low-power on-device ML architectures for autonomous offshore nodes.

Grant Success & Project Growth

All ideas presented here have organically grown from the current project, leading to applications for **4 major EU projects, 2 other external sources, and 1 ETAG PRG** in just the first year.

AQUARIUS
Rejected / Past Submission

BioApps
Rejected / Past Submission

IMAGINE
Rejected / Past Submission

PLANNED COLLABORATIONS

NTNU (Norway) — DAS infrastructure, Svalbard cable (SUBMERSE)
Aalborg University (Denmark) — DAS data processing
KAUST — Red Sea DAS biological benchmarks
Estonian Maritime Museum — Wreck monitoring & historical data

IMPACT

DAS fish chorus detection — proven on Red Sea reefs — transferred to Baltic cod spawning grounds would provide the first **continuous, non-invasive, infrastructure-reusing** monitoring system for a critically endangered EU fish stock.

This project has received funding from the Estonian Research Council under the Mobilitas Pluss programme (Grant no. MOB3JD1212). | MSCA in Estonia, May 2026